

Technology Stack

Date	03 October 2026
Team ID	SWTID-2026-7859
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

The programming languages, frameworks, AI models, tools and APIs used to build EduGenie, with the justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5 (Jinja2 template), CSS3, vanilla JavaScript (Fetch API)	Lightweight, no build step, works on any browser and low-end device; responsive layout with dark mode; JavaScript renders the interactive quiz and formatted results.
2	Backend / Server-Side	Python 3.10+, FastAPI, Uvicorn (ASGI), Pydantic	FastAPI is fast, validates requests with Pydantic and generates interactive API docs (/docs) automatically; Python has the best AI/ML library support.
3	Database / Data Storage	No database (stateless); .env configuration via python-dotenv; Hugging Face cache for local model weights	EduGenie stores no user data, which keeps it simple and private; configuration is kept out of the code.
4	Cloud / Hosting / Deployment	Local deployment with Uvicorn (http://127.0.0.1:8000); Google Cloud (Gemini API) for AI inference	Runs on a laptop for development and demo; cloud AI provides advanced reasoning without needing a GPU. Can be containerized / deployed to Cloud Run later.
5	Version Control & CI/CD	Git, GitHub (https://github.com/Arumparithi42/EduGenie-AI), pytest, VS Code	Version history and project submission on GitHub; pytest suite (40 tests, Gemini mocked) verifies every endpoint; VS Code run/debug/test configurations included.
6	Third-Party APIs / Other Tools	Google Gemini API (google-genai SDK, model gemini-flash-latest); MBZUAI/LaMini-Flan-T5-783M (transformers, torch - optional)	Gemini powers Q&A, summaries, quizzes (JSON mode), learning paths and explanations; LaMini-Flan-T5 gives an offline, CPU-friendly option for concept explanations.